DriveShare: P2P Car Rental

A Scalable Peer-to-Peer Car Sharing Platform Architecture

Final Project Presentation | Software Design Patterns

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The Vision

Simplifying vehicle sharing through a secure, efficient, and user-centric standalone platform. Inspired by Turo, built for high-performance Peer-to-Peer transactions.

Core Capabilities

User Onboarding

Secure registration with 3-question security chain and email authentication.

Asset Management

Owner-led car listings with calendar integration and real-time pricing control.

Smart Booking

Conflict prevention logic ensuring no overlapping rental schedules.

Communication

In-app messaging system facilitating direct owner-renter coordination.

Secure Authentication

Architectural Patterns

- **Singleton:** Manages a unique, global user session.
- **Chain of Responsibility:** Processes password recovery through a sequential validation of security questions.
- **Security:** SHA-256 hashing for all password data.

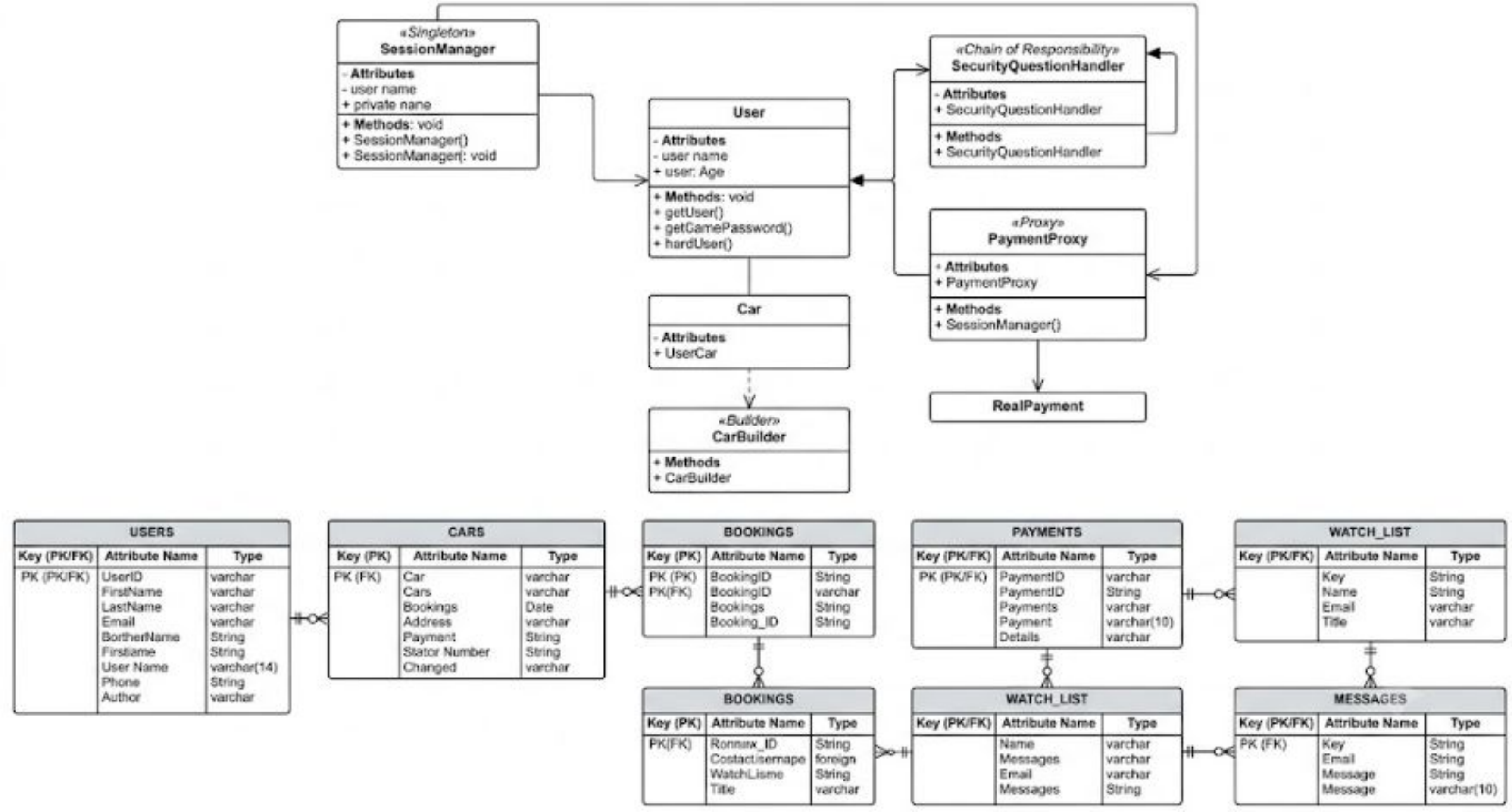
The Owner Suite

Builder Pattern Implementation

Constructs complex car listing objects with optional attributes (mileage, year, pricing) cleanly.

Provides owners with an intuitive dashboard to manage availability calendars and avoid double-booking errors.

UML Diagram


«Singleton»
SessionManager
- Attributes
- user name
+ private nane
+ Methods: void
+ SessionManager()
+ SessionManager(: void
User
- Attributes
- user name
+ user: Age
+ Methods: void
+ getUser()
+ getCamePassword()
+ hardUser()
«Chain of Responsibility»
SecurityQuestionHandler
- Attributes
+ SecurityQuestionHandler
+ Methods
+ SecurityQuestionHandler
«Proxy»
PaymentProxy
- Attributes
+ PaymentProxy
+ Methods
+ SessionManager()
RealPayment
Car
- Attributes
+ UserCar
«Builder»
CarBuilder
+ Methods
+ CarBuilder
USERS
Key (PK/FK) Attribute Name Type
PK (PK/FK) UserID varchar
FirstName varchar
LastName varchar
Email varchar
BortherName String
Firstiame String
User Name varchar(14)
Phone String
Author varchar
CARS
Key (PK) Attribute Name Type
PK (FK) Car varchar
Cars varchar
Bookings Date
Address varchar
Payment String
Stator Number String
Changed varchar
BOOKINGS
Key (PK) Attribute Name Type
PK (PK) BookingID String
PK(FK) BookingID varchar
Bookings String
Booking_ID String
PAYMENTS
Key (PK/FK) Attribute Name Type
PK (PK/FK) PaymentID varchar
PaymentID String
Payments varchar
Payment varchar(10)
Details varchar
WATCH_LIST
Key (PK/FK) Attribute Name Type
Key String
Name String
Email varchar
Title varchar
BOOKINGS
Key (PK) Attribute Name Type
PK(FK) Ronnex_ID String
Costactisernape foreign
WatchLisme String
Title varchar
WATCH_LIST
Key (PK/FK) Attribute Name Type
Name varchar
Messages varchar
Email varchar
Messages String
MESSAGES
Key (PK/FK) Attribute Name Type
PK (FK) Key String
Email String
Message String
Message varchar(10)

Software Demonstration

Will add recording here

Data Infrastructure

Module	Pattern Used	Key Data Elements
Session	Singleton	UserID, Email, Balance
Listings	Builder	CarModel, Year, Price, Calendar
Recovery	Chain of Resp.	Q1/A1, Q2/A2, Q3/A3
Payments	Proxy	BookingID, Amount, TransactionLog

Task Summary

Student	Task(s)	Description
Gavin	Auth, Session Manager, Part of Main, Password Recovery	Developed and deployed core functions of program
Saif	Payments, UI, Models, Messaging, Mediator, Part of Main	Developed and deployed core functions of program
Alexis	UML Diagrams, Slide Deck & Database Dev and Testing	Built project structure, developed database, tested program and pulled together all pieces of project to submit